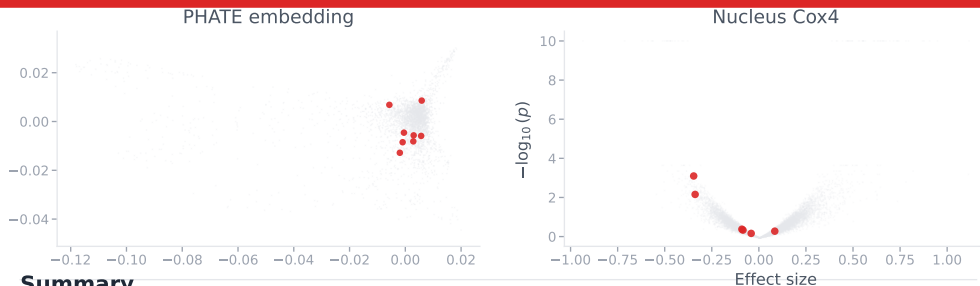


N6-methyladenosine (m6A) RNA methylation

Cluster 221



This cluster shows HIGH confidence for N6-methyladenosine (m6A) RNA methylation as the dominant pathway. Three core components of the WMM complex (VIRMA, WTAP, ZC3H13) are present, representing ~33% of non-control genes. These proteins work together to catalyze m6A methylation of mRNAs, affecting RNA splicing, processing, and stability. The morphological phenotypes observed (ER, nuclear, and cytop...

Established genes

VIRMA, WTAP, ZC3H13

Novel & uncharacterized genes

ADGRF1 novel

G-protein coupled receptor involved in neuronal development and lipid signaling. While GPCRs can influence gene expression programs, a direct mechanistic link to m6A methylation machinery is

PGD novel

Canonical pentose phosphate pathway enzyme producing NADPH. While NADPH is essential for cellular metabolism, a direct link to m6A methylation is not established. May provide metabolic support for

CEP63 novel

Known centrosomal protein involved in centriole duplication and spindle assembly. Recent work shows it stabilizes FXR1, an RNA-binding protein, suggesting potential RNA-related functions. Clustering

SEL1L novel

Well-established role in ER-associated degradation (ERAD) and ER quality control. Its clustering with m6A machinery is unexpected and suggests a potential novel connection between RNA methylation

SPRYD7 unc.

Virtually no functional annotation available. No publications describe its molecular function or biological process. Its clustering with core m6A methylation machinery suggests potential